

Mayank Balmoori

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SUMMARY

Full-stack software engineer with hands-on experience building and shipping production systems across healthcare data infrastructure, real-time collaborative applications, and blockchain/Web3 platforms. Proficient in React, Node.js, TypeScript, WebSockets, and REST API design. Demonstrated ability to deliver end-to-end features independently in fast-paced startup and enterprise environments. Strong foundation in data structures, algorithms, system design, and computer science fundamentals.

EDUCATION

Vellore Institute of Technology (VIT)

Vellore, India

Bachelor of Technology in Computer Science and Engineering

Aug. 2023 – May 2027 (Expected)

- CGPA: 8.97 / 10 Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks, Object-Oriented Programming

EXPERIENCE

Software Engineer Intern

May 2026 – Present

Providence India

Hyderabad, India

- Engineered a resumable large-file transfer system with checkpoint-based failure recovery for healthcare data pipelines, enabling failed transfers to restart from the last committed segment and eliminating redundant full re-transfers on process interruption.
- Designed and implemented a checkpoint persistence layer for cron-triggered file transfer jobs, maintaining transfer state across job executions and process restarts to ensure fault tolerance and reliability at scale.

Full Stack Intern

May 2025 – June 2025

OriginHash

THub, Hyderabad

- Architected and delivered 70% of the Web2 platform within the first month — full-stack development across a React frontend and Node.js/Express backend with MongoDB.
- Designed scalable REST APIs for core data operations; collaborated directly with the founder to drive product requirements and key architectural decisions; shipped end-to-end features from design to deployment independently.

PROJECTS

Canvix [Live Demo](#) [GitHub](#)

Feb. 2026 – Present

React, TypeScript, Canvas API, WebSockets, Node.js, Turborepo

- Built a real-time collaborative whiteboard supporting multi-user drawing, shape manipulation, room-based collaboration, and live synchronization using WebSockets.
- Designed a dual-canvas rendering architecture with separate preview and committed layers, enabling smooth drag, resize, and rotation operations without expensive full-scene re-renders.
- Implemented an event-based undo/redo system tracking shape operations instead of storing canvas snapshots, reducing memory overhead while synchronizing changes across all connected clients in real time.
- Structured as a Turborepo monorepo with shared TypeScript packages and centralized type definitions across the React frontend and Node.js backend; open-sourced on GitHub.

HD Wallet Web App (ETH + SOL) [GitHub](#)

Sept. 2025 – Oct. 2025

Next.js, TypeScript, Ethers.js, Solana Web3.js, BIP39, Blockchain

- Built a multi-chain Hierarchical Deterministic (HD) wallet supporting Ethereum and Solana account generation from a single BIP39 mnemonic phrase.
- Implemented the complete wallet generation pipeline: mnemonic generation, PBKDF2 seed derivation, BIP32 master key creation, and BIP44 chain-specific account derivation.
- Integrated Ethers.js and Solana Web3.js to generate, manage, and display wallet addresses and private keys; documented the build in a technical write-up on [Hashnode](#).

ChatApp [GitHub](#)

Mar. 2025 – May 2025

MERN Stack, WebSockets, Docker, MongoDB

- Developed a real-time chat platform supporting anonymous room-based communication using WebSockets and event-driven message delivery.
- Implemented automatic room cleanup and ephemeral messaging — room state and messages are removed when the final participant disconnects.
- Containerized the full application with Docker for consistent local development and production-ready deployment; backend built with Node.js, Express.js, and MongoDB.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Java, C, C++, SQL

Frontend: React.js, Next.js, Tailwind CSS, Canvas API

Backend: Node.js, Express.js, REST APIs, WebSockets

Databases: PostgreSQL, MongoDB

Cloud & DevOps: Docker, AWS, Git, GitHub, Turborepo, Monorepos

Blockchain: Ethereum, Solana, Ethers.js, Solana Web3.js, BIP39, BIP44

Software Engineering: Agile, OOP, Distributed Systems, API Development, System Design

Computer Science: Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS